

MPs and MCs Project

Home Security System

Concept:

Design a basic home security system using the ATmega32 that detects intrusions and sends alerts via a serial connection.

Components:

- Two ATmega32 microcontrollers
- PIR sensor (for motion detection)
- Magnetic door/window sensor
- Push Button (for arming/disarming the system)
- Buzzer (for local alarm)
- LEDs or 7 Segments (for status)

Functionality:

1. Intrusion Detection:

- PIR sensor: Detects motion within a specific area.
- Magnetic sensor: Detects if a door or window is opened.
- The ATmega32 monitors the state of these sensors through its input pins.
- You can replace PIR sensor and magnetic sensor with switches for simplicity.

2. System States:

- Armed: The system is active and will trigger an alarm if an intrusion is detected.
- Disarmed: The system is inactive and will not trigger an alarm.

3. Push Button Control:

- A push button is used as an external interrupt source to arm and disarm the system.

4. Local Alarm:

- A buzzer connected to an output pin on the ATmega32 sounds an audible alarm when an intrusion is detected while the system is armed.

5. LEDs or 7 Segments:

- The LEDs or 7 segments shows the current system state (armed/disarmed), sensor status.

6. USART Communication:

- The ATmega32 communicates with another ATmega32 via USART.
- This allows for:
 - Remote monitoring of the system state and sensor status.
 - Sending commands to arm/disarm the system or query its status.

Proteus Implementation:

- Simulate the ATmega32, PIR sensor, magnetic sensor, push button, buzzer, LEDs or 7 segments, and USART module in Proteus.
- Implement the control logic, input scanning, and communication protocols in C code for the ATmega32.